S571 S783 S2648 S8754 S669 S461 Ssym Q3421 K3822 K2369 PTMUL PTMUL-D mean of 12

GeoDDG-3D
DDGemb
ESM-MSR (devel) combined
ESM-MSR combined
ESM-MSR (devel) WT head
ESM-MSR WT head
MutateEverything st1
MutateEverything st2
DDMut
SPURS (retrained)
ThermoMPNN + ThermoMPNN-D
ThermoMPNN
ESM3ΔG
SPURS multi (retrained)
IFUM, WT backbone
IFUM, ESMFold
ProteinMPNN (zero-shot)
ESM3-medium (Forge)
ESM3-small (Forge)
ProSST (zero-shot)
ESM3-small-open, σ = 0 (zero-shot)
ESMtherm
ESM3-large (Forge)

0.25 0.50
0.25 0.50 0.75
0.25 0.50 0.75
0.25 0.50 0.75
0.25 0.50
0.25 0.50 0.75
0.25 0.50 0.75
0.25 0.50 0.75
0.25 0.50 0.75
0.25 0.50 0.75
0.0 0.5
0.0 0.5
0.25 0.50 0.75

Spearman ρ (95% protein-level bootstrap); red ring = documented training overlap with the set